Asandei Ștefan-Alexandru

asandei.stefanel@gmail.com | github.com/stefanasandei | linkedin.com/in/stefan-asandei | asandei.com

Education

École Polytechnique Sep 2026 — Present
Bachelor of Science, Mathematics and Computer Science *Anticipated graduation date: June 2029*

Projects

Efficient JPEG Restoration in the Wavelet Domain via Mean Flows arxiv.org/abs/2608.28730
Preprint — Stefan-Alexandru Asandei, Mihai-Alexandru Radu github.com/stefanasandei/jpeg-restore

- Trained a **1-step** diffusion model on LSDIR, optimized for high performance on **consumer GPUs**, measured VRAM usage of **0.805 GiB** and p50 latency of **124.2 ms**, at 1440^2 resolution on an RTX 3090.
- Achieved an LPIPS of 0.1338 at QF20, with **throughput up to 18.82 images/s** at HD resolution
- Conducted ablations on model architectures, flow objectives, and JVP computation algorithms
- Optimized the PyTorch training pipeline by using a memory-efficient JVP (from Nvidia's rCM), reducing peak training memory by **60.9%** while preserving numerical equivalence

Comparison study of PEFT methods github.com/stefanasandei/lora-compose

- Benchmarked 9 PEFT adapters, 4 training recipes, and 6 composition methods on a 1.6B model
- Compared **multi-LoRA merging** techniques, with IterIS achieving 2.9× better subject disentanglement than naive summation.

Competitions

European Olympiad in AI (EUROAI), Silver Medal **Romania**, July 2026

- One of the 6 tasks required auditing a frozen **DistilBERT** sentiment classifier for token-level rationales to identify biases, which I solved with **Integrated Gradients**.
- Another task required identifying redundant parameters in a CNN model, which I solved by thresholding per-weight diagonal **Fisher information** scores to prune low-influence weights while preserving accuracy.

Nitro NLP Hackathon, 1st Place **International (Online)**, April 2026

- The task was to predict total reading time from eye-tracking data. The difficulty was in **out-of-distribution reliability**. My approach used a local validation pipeline, along with analyzing feature importance.
- For feature engineering, I combined **xlm-roberta-large** embeddings with linguistic features (word length, frequency, character-level statistics), for **XGBoost**.

International Olympiad in Artificial Intelligence (IOAI), Bronze Medal **Beijing**, August 2025

- Placed in the **top 30%** globally among 284 competitors representing 61 countries
- Solved 6 tasks over 2 days spanning Computer Vision (CV) and Natural Language Processing (NLP).

Volunteer Work

AI Community Contest, Problem setter, Organizer Sep 2025 — Present

- Organizing monthly contests, to help students better prepare for the international AI Olympiads
- Over 600 community members, from 40+ countries: aicc-official.org
- Relevant published tasks: Find Brain Tumors (Computer Vision, Self-Supervised Learning methods such as BYOL, FixMatch), Watermark Removal (Computer Vision, conditioned Flow Matching training)

European Olympiad in AI, Problem setter January 2026

- Helped with task authoring and review for the 1st practice round
- My task: Star Observatory (Computer Vision, Teacher-Student Knowledge Distillation)

Skills

- **Deep Learning**: PyTorch, OpenCV, NumPy, Transformers, Diffusers, Weights & Biases, Hydra
- **Programming Languages**: Python, C++, Bash
- **Technologies**: Docker, vLLM, PostgreSQL, Redis
- **Languages**: Romanian (Native), English (C2 Proficient)
- **Soft Skills**: Problem Solving, Collaboration, Adaptability, Leadership